QP Code:……. **Reg. No. ……**

Name:…….

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

THIRD SEMESTER MGU – UGP (HONOURS) REGULAR

EXAMINATION OCTOBER 2025

Third Semester (Model Question Paper)

Discipline Specific Elective – MG3DSECHE200 - Basic Analytical Chemistry

(2024 Admission Onwards)

Duration: 2 Hrs. Maximum Marks: 70

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain] [CO No(s)]

Part A

Short Answer Questions

Answer any ***five*** questions.

Each questions carries 4 marks

1. What are the importance of mean and variance in the statistical analysis of data? [U] [1]
2. Explain the role of ionic product and solubility product in the inter-group separation of cations in qualitative inorganic analysis. [U] [1]
3. Briefly discuss the importance and advantages of maintaining a laboratory notebook as a safe laboratory practice. [A] [2]
4. How do you carryout the calibration of a standard flask? [K] [2]
5. What is a redox reaction? Explain the chemistry of $KMnO_4$ reacting with ferrous ion. [A] [3]
6. Write down the steps involved in the gravimetric estimation of barium as barium sulphate. [K] [3]
7. Explain the basic concepts of chromatography. Which are the most widely used chromatographic techniques in analysis of samples? [A] [4]

(5 x 4 = 20)

Part B

Short Essay Questions

Answer any ***five*** questions.

Each questions carries 7 marks

8. Briefly discuss the basic principles of regression analysis. [U] [1]
9. What are determinate and indeterminate errors? How would you reduce the determinate errors in an experiment? [U] [1]
10. Describe the safety precautions you must take while working in a chemical laboratory. [K] [2]
11. What are interfering anions? Explain how borate and phosphate interfere in the analysis of cations via inter – group separation. [U] [1]
12. Which is the indicator used in the titration of EDTA against Zn^{2+} ions? Explain the principle of indicator action in this titration. [U] [3]
13. Explain how recrystallization and precipitation help in separation and purification of components of mixtures. [K] [4]
14. Briefly discuss the basic principles and applications of HPLC and gas chromatography. [A] [4]

(5 x 7 = 35)

Part C

Essay Questions

Answer any ***one*** questions.

Each questions carries 15 marks

15. Explain various types of conductometric titrations of acids and bases with suitable examples. [U] [3]
16. Discuss the basic principles of separation techniques (a) fractional distillation (b) solvent extraction. [A] [4]

(1 x 15 = 15)

END OF THE QUESTION PAPER

***